



ML - previous question paper

Machine Learning (APJ Abdul Kalam Technological University)



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APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY
B.Tech Degree S7 (R, S) Examination November 2024 (2019 Scheme)

Course Code: CST413

Course Name: MACHINE LEARNING

Duration: 3 Hours

Max. Marks: 100

PART A

Marks

Answer all questions, each carries 3 marks.

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|----|---|-----|
| 1 | How is machine learning different from traditional programming? | (3) |
| 2 | Identify and explain the suitable learning method in each case | (3) |
| | a) Analyze social media posts to identify sentiment (positive, negative, neutral) without prior sentiment labels. | |
| | b) Develop an intelligent agent that plays a video game and learns to maximize its score through trial and error. | |
| 3 | Compare and contrast linear regression and logistic regression. | (3) |
| 4 | Explain how the Gradient Descent algorithm updates model parameters during training | (3) |
| 5 | Calculate the output of a 3-input neuron with inputs $X = [0.5, 1.5, 0.8]$, weights $W = [0.7, 0.4, 0.6]$ and bias $b = -0.1$. Use the <i>tanh</i> activation function. | (3) |
| 6 | Why can the Perceptron solve some Boolean functions, like AND and OR, but not XOR? | (3) |
| 7 | Write short notes on single, complete and average linkage criteria used in hierarchical clustering. | (3) |
| 8 | Explain the significance of the expectation and maximization steps in the Expectation-Maximization (EM) Algorithm. | (3) |
| 9 | Why is cross-validation a valuable technique in preventing overfitting? | (3) |
| 10 | What are the characteristics of a model that has high bias and low variance? | (3) |

PART B*Answer any one full question from each module, each carries 14 marks.***Module I**

- 11 a) Find the Maximum Likelihood Estimation for the parameter p in the binomial distribution whose probability function is $f(x) = \binom{n}{x} p^x 1 - p^{n-x}$, (8)

$$x = 0, 1, 2, \dots, n$$

- b) Compare and contrast Maximum Likelihood Estimation (MLE) and Maximum A Posteriori (MAP) estimation. (6)

OR

- 12 a) Consider a geometric distribution with the probability mass function $P(X = k) = \theta(1 - \theta)^{k-1}$. Suppose n i.i.d. samples are drawn from this distribution. (8)
- Derive an expression for the log-likelihood of the data as a function of the parameter θ .
 - Determine the maximum likelihood estimate (MLE) for θ .
 - Assume θ has a Beta prior distribution. What is the posterior distribution of θ ?
- b) Given a feature vector instance X and a class label c_k , describe how Bayes' theorem is used to calculate the probability $P(c_k | X)$. (6)

Module II

- 13 a) Consider the following dataset S , which contains information about customers applying for a loan. The target attribute is $Loan_Approval$, which indicates whether a loan was approved (Yes / No). (8)

Customer ID	Age_Group	Income_Level	Credit_Score	Loan_Approval
1	Young	High	Good	Yes
2	Middle	Medium	Fair	No
3	Old	Low	Poor	No
4	Young	Medium	Good	Yes
5	Middle	High	Good	Yes
6	Old	Medium	Fair	No
7	Young	Low	Poor	No
8	Middle	Medium	Good	Yes

- Calculate the entropy of the dataset S with respect to the target attribute $Loan_Approval$.
 - Identify the first splitting attribute for a decision tree by using ID3 algorithm.
- b) Differentiate between Lasso and Ridge regularization methods. (6)

OR

- 14 a) A teacher wants to predict students' final exam scores based on the number of hours they studied. She collects the following data for five students. (8)

Student	Hours Studied	Final Exam Score
1	2	75
2	3	85
3	1	65
4	4	90
5	5	95

- Fit a linear regression model to predict final exam scores based on hours studied. Write down the regression equation.
- Use the model to predict the final exam score for a student who studied for 3.5 hours.
- Interpret the slope coefficient for hours studied in the context of this analysis.

- 15 b) Explain how Naive Bayes handles multi-class classification problems. How does the model calculate probabilities for multiple classes, and how is the final prediction chosen? (6)

Module III

- 15 a) Describe the backpropagation process in neural network training. How does this algorithm adjust the weights to reduce the error in the model's predictions? (8)
- b) Consider a linear support vector machine (SVM) used for binary classification. If you take a correctly classified data point that is situated far from the decision boundary and remove it from the training dataset, what will happen to the decision boundary when you retrain the classifier? Will it change or stay the same? Provide a justification for your answer. (6)

OR

- 16 a) Describe the mathematical formulation of the SVM optimization problem and outline the procedure for solving it. (8)
- b) Explain the concept of "leaky ReLU" and why it was introduced. What problem does leaky ReLU aim to solve, and how does it differ from standard ReLU? (6)

Module IV

- 17 a) Consider the following set of points: $P = (4, 1)$, $Q = (6, 3)$, $R = (2, 5)$, and $S = (7, 7)$. Use agglomerative hierarchical clustering with the complete linkage method (8)

to construct the clustering step-by-step. Create the corresponding dendrogram to visualize the process.

- b) Consider the tuples (12, 8, 3, 5) and (7, 4, 10, 2): (6)
- Calculate the Euclidean distance between the two tuples
 - Calculate the Manhattan distance between the two tuples
 - Calculate the Minkowski distance with $p = 3$ between the two tuples

OR

- 18 a) Consider the following data points: $A1 = (1, 2)$, $A2 = (4, 6)$, $A3 = (2, 5)$, $A4 = (9, 1)$, and $A5 = (7, 3)$. Perform two iterations of the K-Means clustering algorithm with $k = 2$ and give the resulting clusters and centroids. Use the initial centroids as $A1$ and $A4$. (8)
- b) Explain the procedure for implementing Principal Component Analysis (PCA) to achieve dimensionality reduction, and analyze the advantages and disadvantages associated with its use. (6)

Module V

- 19 a) You have a confusion matrix from a spam detection model: (8)

	Predicted Spam	Predicted Not Spam
Actual Spam	80	20
Actual Not Spam	15	85

Calculate the precision, recall, accuracy, and F1 score for this model. Interpret the results of these metrics in the context of the spam detection task.

- b) Explain the concept of bootstrapping in the context of model evaluation. How does bootstrapping differ from traditional cross-validation techniques? (6)

OR

- 20 a) What is the Area Under the Curve (AUC) in relation to the Receiver Operating Characteristic (ROC) curve? How does AUC help in comparing different classification models? (8)
- b) What are ensemble methods in machine learning? Explain the difference between bagging and boosting as ensemble techniques. (6)
